

## Section 4.3: The different Types of Networks

There are different **types** of **computer networks** and they all have their **advantages** and **disadvantages**.

In this section we will discuss the features each type.

### Key Concepts of this section:

- # Understand that there are **four types** of computer networks.
- # Know the **features** of each type of computer network.
- # Be able to **describe** the **advantages/disadvantages** of each type of computer network.

## Types of computer networks

# There are four main types of computer networks:

- **LAN** (local area networks)
- **WLAN** (wireless local area networks)
- **WAN** (wide area networks)
- **Bluetooth**.

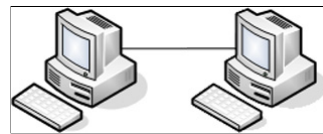
We will discuss the **features** of each network type, as well as their **advantages/disadvantages** below:



### Key Words:

LAN, WLAN, WAN, Bluetooth.

### Examples:



A network is just two or more computers joined together.

## 1 Local Area Networks (LAN)

# Local area networks (or **LAN's**) are usually located in a **single room** or **small building**.

LAN's are computer networks that are **confined** to **small areas**.

# An example of a LAN would be a **school network**. The computers on the network would be confined to a classroom or the school.

# LAN's are **private** and can only be accessed by people in the room or the building.

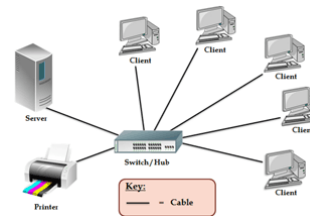
### For example:

A school network is only used by students who attend the school.

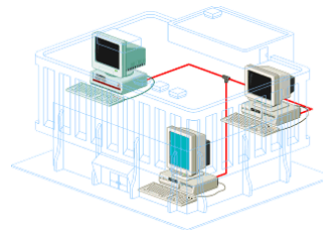
# A typical LAN would consist of **several computers** that are **connected** to each other and can share resources such as **printers** and **scanners**.

# Computers and devices on a LAN are connected using **cables**.

### Examples:



A LAN consists of several computers connected together in order to share resources. (click to zoom)



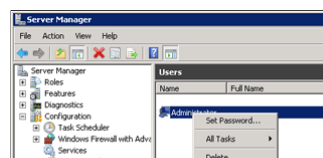
LAN's are confined to one room or a small building.



LAN's allow networked computers to share resources (like printers).

### Advantages and disadvantages of LAN's:

| Advantages of LAN's                                                                                                                                                                               | Disadvantages of LAN's                                                                                                                                                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Saves money</b> as each computer on the LAN can share resources.<br><br>(For example - you only need to buy one printer because it can be shared between all the computers on the network)     | <b>Viruses</b> can <b>spread</b> around LAN's very <b>quickly</b> .<br><br>(This is because the computers are all joined together - if one computer gets infected, the other are at risk) |
| Files and data can be <b>shared</b> easily.<br><br>(Like the shared drive on a school network)                                                                                                    | <b>Security</b> can be an <b>issue</b> .<br><br>(If one computer is hacked into, the other computers on the network can also be accessed)                                                 |
| Files and data can be <b>accessed</b> from <b>any computer</b> on the network.<br><br>(For example - it doesn't matter which computer in the school you log onto, you can access your work files) | The network can become <b>unusable</b> if the main <b>server</b> computer <b>breaks down</b> .<br><br>(You won't be able to log onto any of the client computers on the network)          |
| Data (like <b>emails</b> ) can be sent around a LAN <b>very fast</b> .<br><br>(The cables that connect the computers can handle very fast data transfer)                                          | <b>Slow</b> internet connections.<br><br>(All of the computers on the network share the same internet connection. This can make it slow for users)                                        |
| LAN's can be <b>monitored</b> easily.<br><br>(Network managers can check people's                                                                                                                 | Initial <b>cost</b> of installing the LAN is <b>high</b> .<br><br>(Buying cables and equipment etc)                                                                                       |



internet usage or change passwords etc)



LAN's allow network managers to monitor users (like checking internet usage and passwords).

## 2 Wireless Local Area Networks (WLAN)

- # WLAN's are similar to LAN's but they **connect computers** together **wirelessly** instead of using cables

This is achieved using **Wi-Fi** technology.

- # **Routers** or **wireless switches** are used to **send** and **receive** data around the network.
- # Client computers or devices send and receive data using **wireless adaptors**.

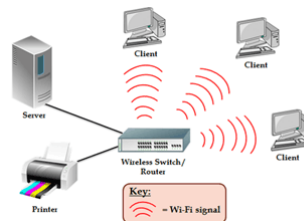
Most modern **laptops**, **tablets** and **mobile phones** come with **built-in Wi-Fi** adaptors.

- # A typical router can send/receive data to/from computers up to **50 metres** away.

### Note:

Most **home networks** are **WLAN's**. They use a router to allow several computers or devices to use the same internet connection.

### Examples:



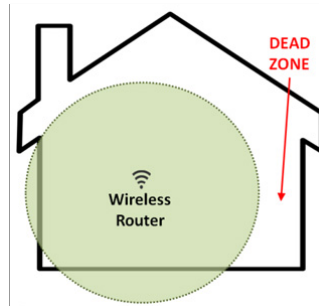
WLAN's use wireless technology instead of cables to connect computers and devices to the network. (click to zoom)



A router can be used to send/receive data wirelessly to/from devices on the network.

### Advantages and disadvantages of WLAN's:

| Advantages of WLAN's                                                                                                                                                      | Disadvantages of WLAN's                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Physically safer</b> than a LAN because there are no cables to trip over.                                                                                              | Some areas in the building <b>may not pick up</b> the wireless signal.<br><br>(This means that computers or devices may not connect to the network from certain locations)                    |
| <b>Cheaper</b> to set up than a WLAN because you don't have to buy lots of expensive cables.                                                                              | <b>Security</b> can be an <b>issue</b> as anyone with a wireless enabled device can try to access the network.<br><br>(It's very important the network is secured with an encrypted password) |
| <b>Easier</b> to <b>add more computers</b> to the network.<br><br>(All they need is a wireless adaptor and they can then communicate with other computers on the network) | Wireless signals can be <b>unreliable</b> as they sometimes suffer from <b>interference</b> from other electronic equipment.<br><br>(Cables used on LAN's are far more reliable)              |
| Devices such as <b>printers</b> and <b>scanners</b> can be <b>connected</b> to the network easily.<br><br>(Some printers/scanners have built-in wireless adaptors)        | <b>Data transfer</b> around a wireless network is much <b>slower</b> compared to a <b>LAN</b> .                                                                                               |
| <b>Computers</b> and <b>devices</b> can be used <b>anywhere</b> in the building.<br><br>(As long as they are in range of the network's router)                            | WLAN's can be <b>useless</b> for use in a <b>large building</b> as the signal range is limited.<br><br>(Computers can typically communicate with the router up to about 50 metres)            |



Devices outside of the router's wireless range limit will not be able to connect to the network.



Any wi-fi enabled devices can connect to a wireless network. (click to zoom)

## 3 Wide Area Networks (WAN)

- # Wide area networks are made up of **computers** and **devices** that are connected over a **large geographical area**.

The **internet** is the world's **largest WAN**.

The internet is just **millions of computers** all over the world that have been connected together so they can **share data**.

- # WAN's are created by **connecting** lots of **LAN's** and individual **computers** together

### Examples:



WAN's are created by connecting lots of LAN's and individual computers together.

Every time you switch on your **Wi-Fi connection** or access **mobile data** (on your phone), you are connecting your device or computer to a WAN.

# Computers and devices can connect to a WAN using **telephone lines**, **fibre-optic cables** and **satellite signals**.

# In order to connect your computer to a WAN you also need a **router** or a **modem**.

Routers and modems allow your computer to **send/receive data to/from** other computers on the WAN.



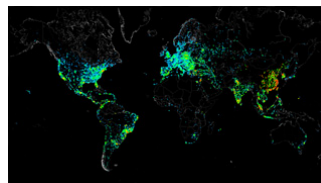
WAN's are lots of networks and computers connected together over a large area. (click to zoom)



Computers and devices can connect to a WAN using routers, cables or even satellite signals.

### Advantages and disadvantages of WAN's:

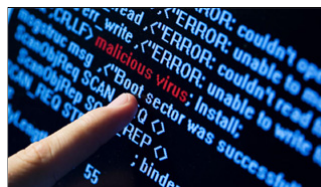
| Advantages of WAN's                                                                                                                                                                                          | Disadvantages of WAN's                                                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Computers can be <b>connected</b> over <b>wide areas</b> .<br>(Across cities or even continents)                                                                                                             | <b>Security</b> can be an <b>issue</b> as anyone with access to the internet can potentially access any of the computers on the network.<br>(Computers on the network need to be secured with a firewall and important files should be encrypted) |
| <b>Files</b> and <b>data</b> can be <b>shared</b> over a <b>large area</b> .                                                                                                                                 | It's very <b>easy</b> to accidentally <b>download viruses</b> from a WAN onto your computer.<br>(You need to make sure that your computer is protected with up-to-date anti-virus)                                                                |
| People can use their computers/devices to <b>communicate</b> very quickly, over large areas.<br>(Sending emails, discussion forums, video conferencing etc.)                                                 | Data is <b>transferred</b> across a WAN at a much <b>slower rate</b> than it is across a LAN.<br>(Download speeds are limited)                                                                                                                    |
| <b>E-Commerce</b> (shopping) websites can be set up and accessed by people from all over the world.<br>(This allows people with disabilities to purchase items online and have them delivered to their door) | <b>Monitoring</b> a WAN can be <b>difficult</b> because they have so many computers connected to them.<br>(This means that inappropriate content may be uploaded)                                                                                 |
| Computers and devices can be used <b>anywhere</b> in the building.<br>(As long as they are in range of the network's router)                                                                                 |                                                                                                                                                                                                                                                   |



The internet is the world's largest WAN. It is millions of computers and devices linked together all over the world.



Transferring files across a WAN can be slow.



Computers connected to a WAN are more at risk of security risks like viruses and hackers.

## 4 Bluetooth (WPAN)

# Bluetooth is also known as a **Wireless Personal Area Network (WPAN)**.

# Bluetooth devices can connect to each other from between **1 and 100 metres** away.

The connection is made **without** using **wires** or **cables**.

Bluetooth uses **radio waves** to transmit data.

# Once two devices are connected with bluetooth, you can **transfer files** and **data** between them up to the maximum range of **100 metres**.

#### Note:

Devices which can make use of bluetooth include:



Mobile phones



Tablets



Wireless Mouse



Wireless Speakers

#### Examples:



Bluetooth networks are known as 'personal networks' because devices have to be close together in order to connect. (click to zoom)



Bluetooth devices connect together wirelessly up to a range of 10 metres.

### Advantages and disadvantages of Bluetooth:

| Advantages of Bluetooth | Disadvantages of Bluetooth |
|-------------------------|----------------------------|
|                         |                            |

|                                                                                                                                            |                                                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Very <b>easy</b> to <b>connect</b> bluetooth devices together.<br>(you just switch on your bluetooth and wait for the devices to pair)     | <b>Limited</b> signal range.<br>(With the normal signal range limit of only 10 metres, the two devices need to be close-by to connect)                                           |
| Bluetooth signals are <b>not easily</b> to <b>disrupt</b> .<br>(They aren't easily affected by other electrical devices like Wi-Fi can be) | <b>Slow</b> data transfer. Bluetooth is not suitable for transferring large files.<br>(Bluetooth cannot transfer data between devices as fast as Wi-Fi.)                         |
| Works <b>wirelessly</b> so there are <b>no</b> messy <b>cables</b> needed to connect devices.                                              | <b>Very unsecure</b> connections and easy for hackers to break into.<br>(Bluetooth is not as secure as Wi-Fi and makes it unsuitable for transferring sensitive or private data) |
| Bluetooth uses <b>very little power</b> .<br>(This means that using bluetooth won't drain your devices battery quickly)                    |                                                                                                                                                                                  |



Bluetooth transfer speeds are very slow.



Bluetooth connections are not very secure and easy for hackers to break into.

**Activity!**[Different Types of Networks - Research Sheet](#)

Click the above task and answer the questions about the 4 different types of computer networks.

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